Notes on Extracting Text from PDF Files

1 Why order matters

A PDF file describes where each character is drawn on the page. It does not record which sentence a character belongs to, or the order in which a person should read the text. Extraction tools therefore have to rebuild that order from the positions alone.

The simplest tools skip that work and return characters in the order they are stored in the file. When the program that wrote the file stored them in reading order, the result looks correct.

2 When it breaks

Some programs write a two-column page one visual row at a time. The first line of the left column is followed by the first line of the right column, and so on down the page.

Read in stored order, every sentence is cut in half and joined to an unrelated sentence from the other column. The output still contains real words and looks like valid prose at a glance.

That is what makes the problem dangerous for a research agent. A missing page is easy to notice, but a page of plausible, mixed sentences is quoted and cited as if it were the source.

3 What a fix needs

A layout-aware extractor groups characters into blocks, detects the columns, and reads each column from top to bottom before moving to the next one. That restores the intended order.

Detecting the problem is cheaper than fixing it: compare the stored order of the text blocks with their positions on the page, and only send the pages that disagree to the costly path.